

TRAVEL PLANNER AGENT

AI-Powered Intelligent & Personalized Travel Planning System

1. Problem Statement

The Challenge – A Travel Planner Agent is an AI-powered assistant that helps users plan trips efficiently and intelligently. It uses real-time data to suggest destinations, build itineraries, and recommend transport and accommodation options. By understanding user preferences, budgets, and constraints, it tailors personalized travel plans. Integrated with maps, weather updates, and local guides, it ensures a smooth travel experience. The agent can also manage bookings, alert users to changes, and optimize schedules on the go. This smart assistant transforms complex travel planning into a seamless, enjoyable process.

Mandatory Requirement: Use of IBM Cloud Lite services / IBM Granite is mandatory.

2. Proposed Solution

The proposed solution is an **AI-powered Travel Planner Agent** built using IBM Granite and IBM Cloud Lite services. It provides a single intelligent platform where users enter their destination, travel dates, budget, number of travelers, interests, and preferences. The system then generates a personalized, optimized, and adaptable travel plan.

- **Personalized Trip Planning** – Recommends destinations, attractions, activities, transport, and accommodation according to the user's preferences and budget.
- **Intelligent Itinerary Generation** – Creates a day-wise itinerary considering distance, travel time, opening hours, activities, and available time.
- **Real-Time Travel Information** – Uses weather, maps, transport, and destination information to provide relevant and updated recommendations.
- **Budget Management** – Estimates transportation, accommodation, food, and activity expenses and helps users stay within their selected budget.
- **Dynamic Optimization** – Updates the itinerary when weather, schedules, availability, or user requirements change.
- **Conversational AI Assistant** – Allows users to modify plans naturally, such as "Make Day 2 cheaper" or "Replace the outdoor activity because of rain."

3. Technology Stack

Layer	Technology	Purpose
AI / LLM	IBM Granite	AI reasoning, natural-language understanding and itinerary generation
Cloud	IBM Cloud Lite	Cloud deployment and supporting infrastructure
Frontend	React.js	Interactive travel dashboard and chatbot interface
Backend	Node.js + Express.js	APIs, business logic and agent orchestration
Database	MongoDB	User preferences, itineraries and travel data
RAG	Vector Database / Knowledge Base	Retrieval of relevant and reliable travel information
External APIs	Maps / Weather / Travel APIs	Real-time routes, weather and travel information
Version Control	Git + GitHub	Source-code management and collaboration